

ScavengeAR: From Licensing Fees to Free — Rebuilding Mobile AR with Unity-Native Tools

Victor Vo Leung
Independent Developer
Pleasanton, California, USA
thevictor2225@gmail.com

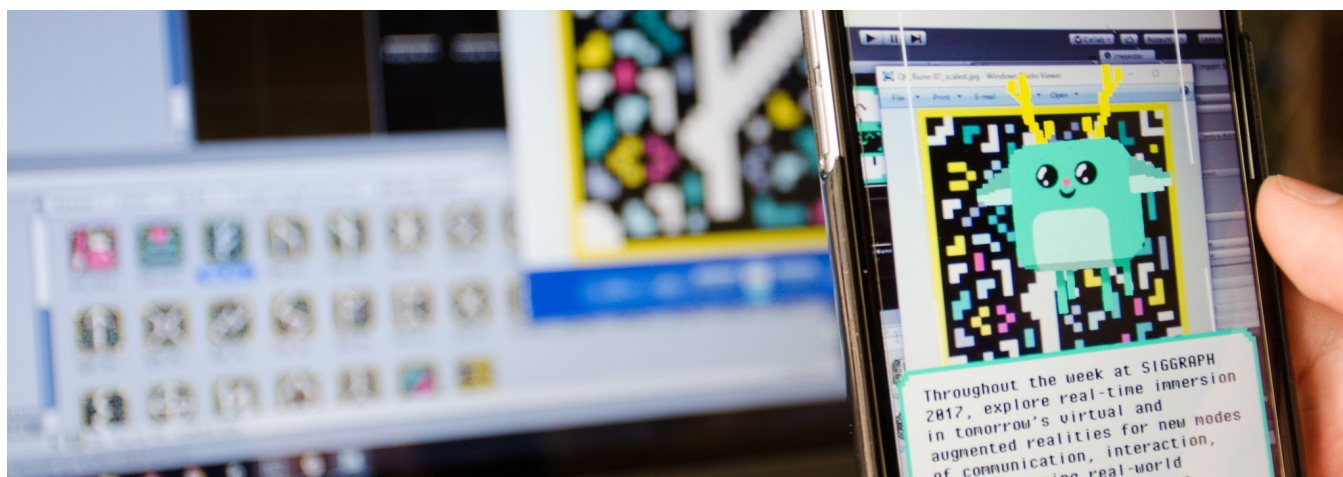


Figure 1: Testing ScavengeAR's 2017 mobile build in Unity

Abstract

ScavengeAR, an augmented reality creature-collecting mobile game, returns to SIGGRAPH in 2025 after a six-year hiatus, featuring a modernized tech stack and refined gameplay. The reboot preserves the core player experience while eliminating costly third-party dependencies.

ACM Reference Format:

Victor Vo Leung. 2025. ScavengeAR: From Licensing Fees to Free — Rebuilding Mobile AR with Unity-Native Tools. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Appy Hour (SIGGRAPH Appy Hour '25)*, August 10–14, 2025, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3721260.3733980>

1 Introduction

ScavengeAR was SIGGRAPH's official AR mobile app from 2017 to 2019, offering attendees an engaging augmented reality experience and providing Luddy School of Informatics, Computing, and Engineering, Indiana University Indianapolis students with hands-on opportunities in interactive media development.

In the game, SIGGRAPH attendees choose a role — Artist, Scientist, or Educator — and explore the conference venue to discover

printed image targets that, when scanned through the app, spawn 3D AR creatures into their physical surroundings. Like a digital safari, players use the in-app camera to photograph each creature, capturing them into a personal library. New creatures are released each day of the conference, encouraging daily exploration and unlocking new layers of narrative and prize opportunities over time. In addition, the attendee has their own creature based on their role, which encourages real interactions with other attendees in order to catch all the creatures for the best prizes.

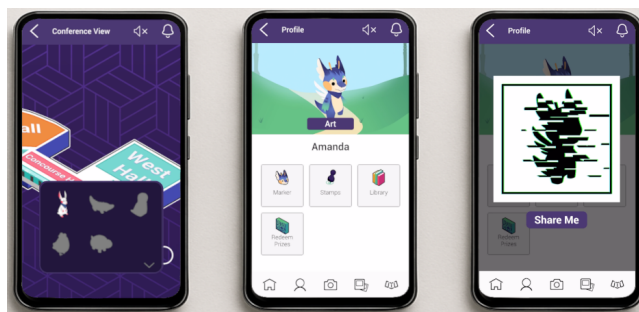


Figure 2: ScavengeAR 2019 Los Angeles Convention Center map screen, profile screen, and role share screen.

2 Motivation

ScavengeAR ran from 2017-2019, with over a thousand daily active users during the conference. In 2020, the Covid Pandemic stopped

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SIGGRAPH Appy Hour '25, Vancouver, BC, Canada
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1552-5/25/08
<https://doi.org/10.1145/3721260.3733980>

in-person activity for a number of years, and effectively halted the app's development, leaving it in a deprecated state.

In 2025, a small volunteer team undertook a comprehensive refactoring of ScavengeAR to improve its maintainability and accessibility, in order to revive the spirit of the app to a modern audience.

3 Technical Approach

The original version of ScavengeAR was built using Unity 2018 and Vuforia 8 - a robust and accessible image tracking solution that was among the most practical options for AR based 2D markers at the time. Vuforia's artist-friendly pipeline utilized a parent-child GameObject hierarchy within a single-scene architecture, allowing low-code AR development and rapid prototyping in a pre-ARKit/ARCore standardization era. However, Vuforia also required licensing fees for production deployment and cloud-hosted image target storage.

For the 2025 relaunch, ScavengeAR was refactored to using Unity 2022 and AR Foundation 5 - Unity's native, cross-platform AR framework built on top of ARKit and ARCore - to support lightweight, offline 2D image tracking and reduce long-term costs, build size, and technical dependencies.

Migrating to AR Foundation required a shift to its modular, multi-scene, prefab-driven architecture. While we traded Vuforia's simplicity, the new system offered improved performance management and a clearer separation of concerns. AR Foundation also supports full in-editor XR simulation—replacing webcam-based workflows—which proved essential for rapidly iterating on the modernized photo capture experience.

Originally built using GameSparks for backend services and a licensed data binding plugin for UI management, we replaced GameSparks with Unity's built-in serialization for offline data handling, and transitioned to an open-source MVVM Toolkit. These changes not only further reduced external dependencies and licensing costs but also simplified the codebase, making it more approachable for new contributors.

While the core functionality remains consistent with previous iterations, this overhaul ensures that ScavengeAR can continue to be a sustainable and educational project. By leveraging Unity's modern XR capabilities and open-source tools, we've positioned the app for easier updates and potential future enhancements, aligning with our goal of providing valuable learning experiences for students and enjoyable interactions for conference attendees.

4 Art and Design



Figure 3: New creatures for ScavengeAR

Luddy School of Informatics, Computing, and Engineering, Indiana University Indianapolis students has historically contributed original 3D creature designs for each iteration of ScavengeAR, and they return once again with a new cast of characters. This year's

creatures were developed using a traditional animation and visual effects pipeline, continuing the project's tradition of experimenting with new styles and creative direction in each version.

For the user experience, we retained the core interaction flow from ScavengeAR 2019, which has been iteratively tested and refined through multiple conference deployments. While earlier versions of the app featured playful mini-games like Triviatron and Sigglet Falls, these proved non-essential to the primary gameplay loop. Instead, our focus shifted to improving the AR photo-taking experience—updating the camera overlay to better suit modern tall-screen phones, adding a tactile shutter animation, and introducing a polaroid-style fadeout to reinforce the analog camera motif.

5 Future Work



Figure 4: A 2018 ScavengeAR creature, originally featured at the SIGGRAPH Studio venue in the Vancouver Convention Centre, shown here without the mobile user interface.

Mobile AR has historically offered a more accurate representation of augmented reality by leveraging the raw camera feed to detect physical markers in the environment. At the time, Mixed Reality headsets restricted access to raw passthrough video, limiting similar capabilities. That trend is now shifting, with newer headsets offering improved access to video passthrough.

As we look ahead, a key challenge will be unifying the user experience (UX), user interface (UI), and interaction patterns across both mobile and headset-based platforms. With our latest refactor, we're well-positioned to support both moving forward.

Acknowledgments

ScavengeAR could not have been created without the significant contributions of: Casey Kwock, Victor Leung, Thach Nguyen, Louie Whitesel, and Zeb Wood. Additionally, we appreciate the supporting contributions of Jose Garrido, Spencer Hayes, Alli Johnson, and Andy Wang.

We also thank the many volunteers who helped bring previous versions of the game to life, as well as the donors whose generosity helped cover software licensing costs throughout the project's development. Finally, we extend our gratitude to the players across SIGGRAPH conferences whose enthusiasm and feedback have continually shaped the experience.